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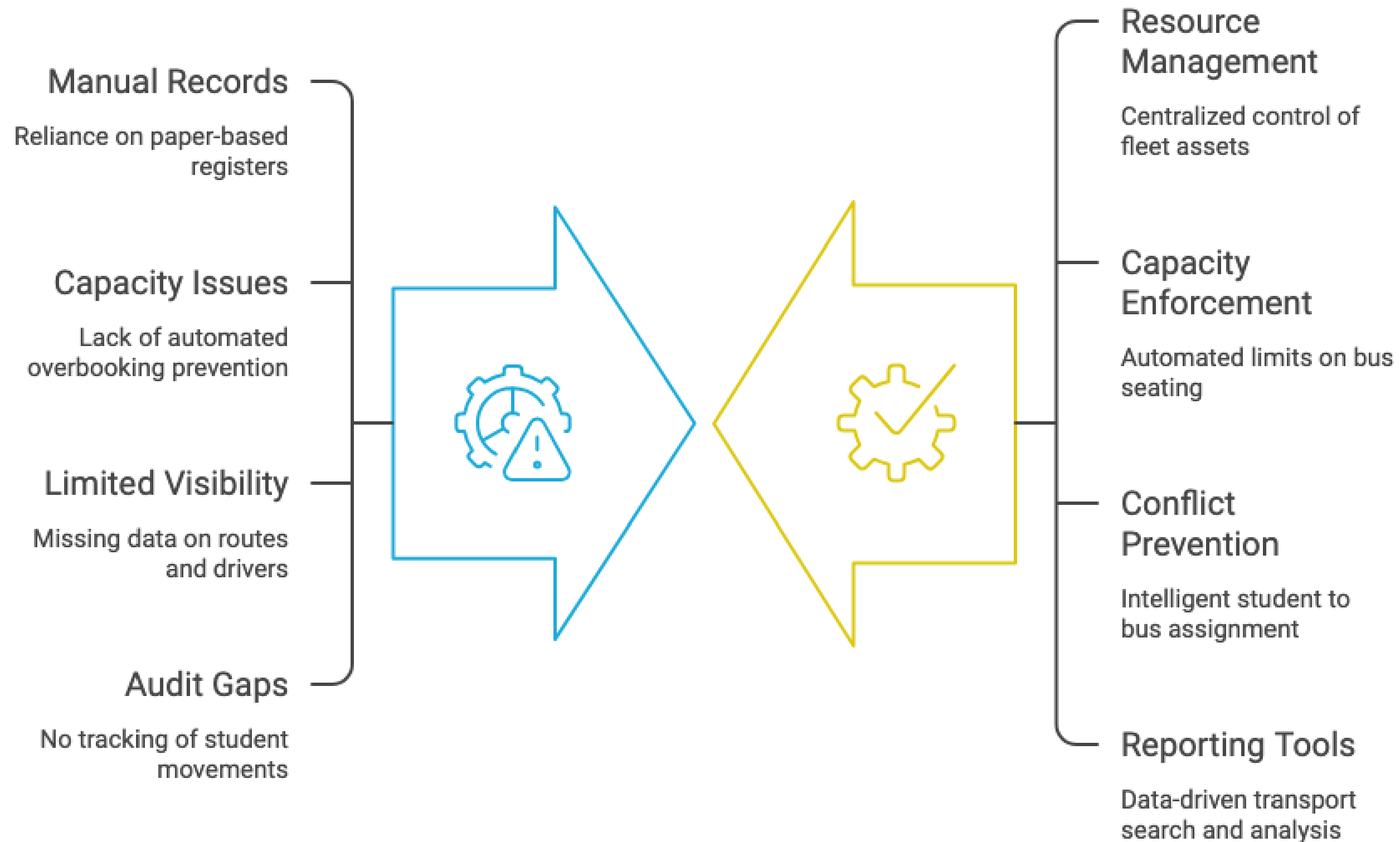
JAVA Programming

School Bus Tracking & Management System

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Problem & Objective

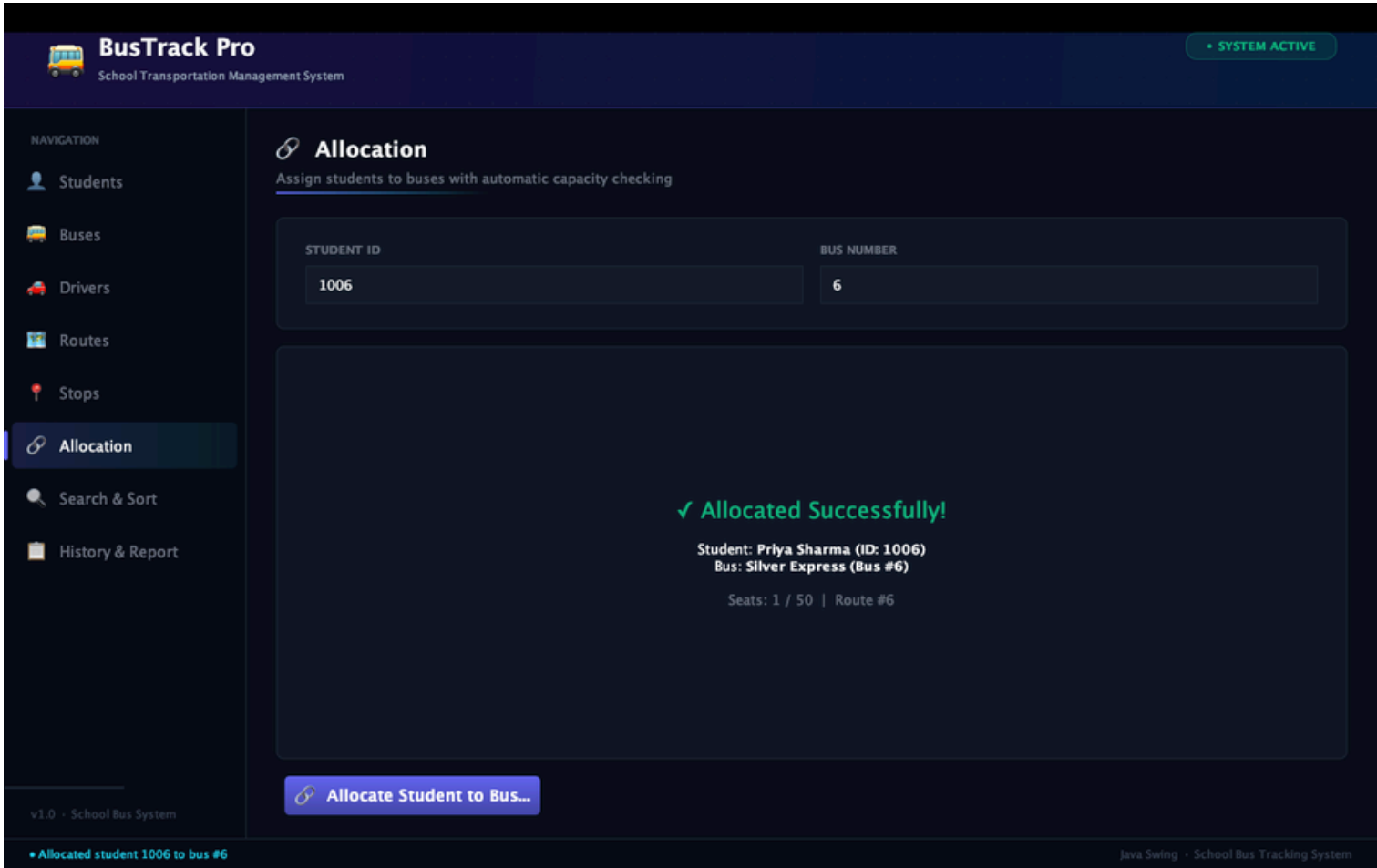
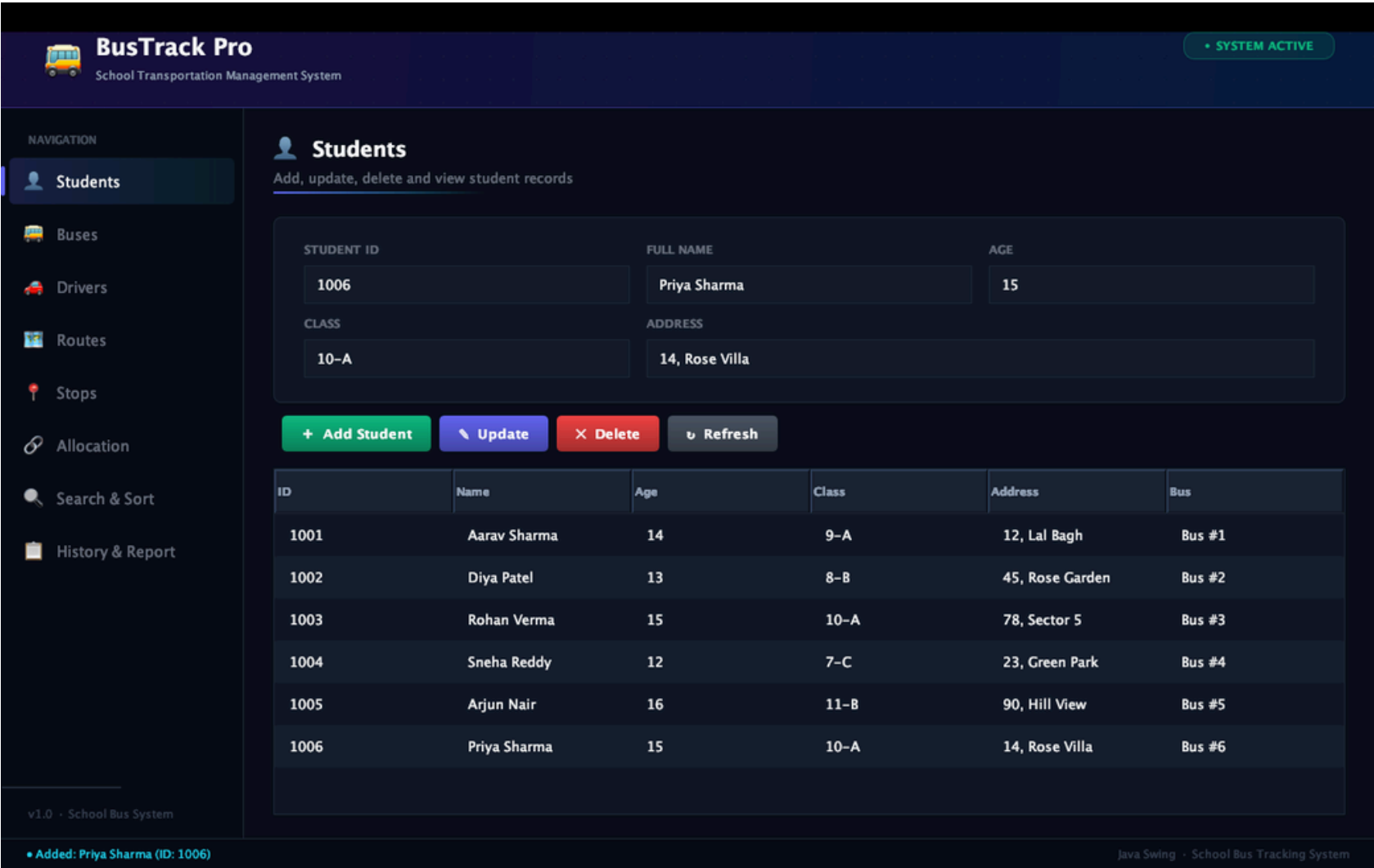
Optimized School Transport Management



System Architecture

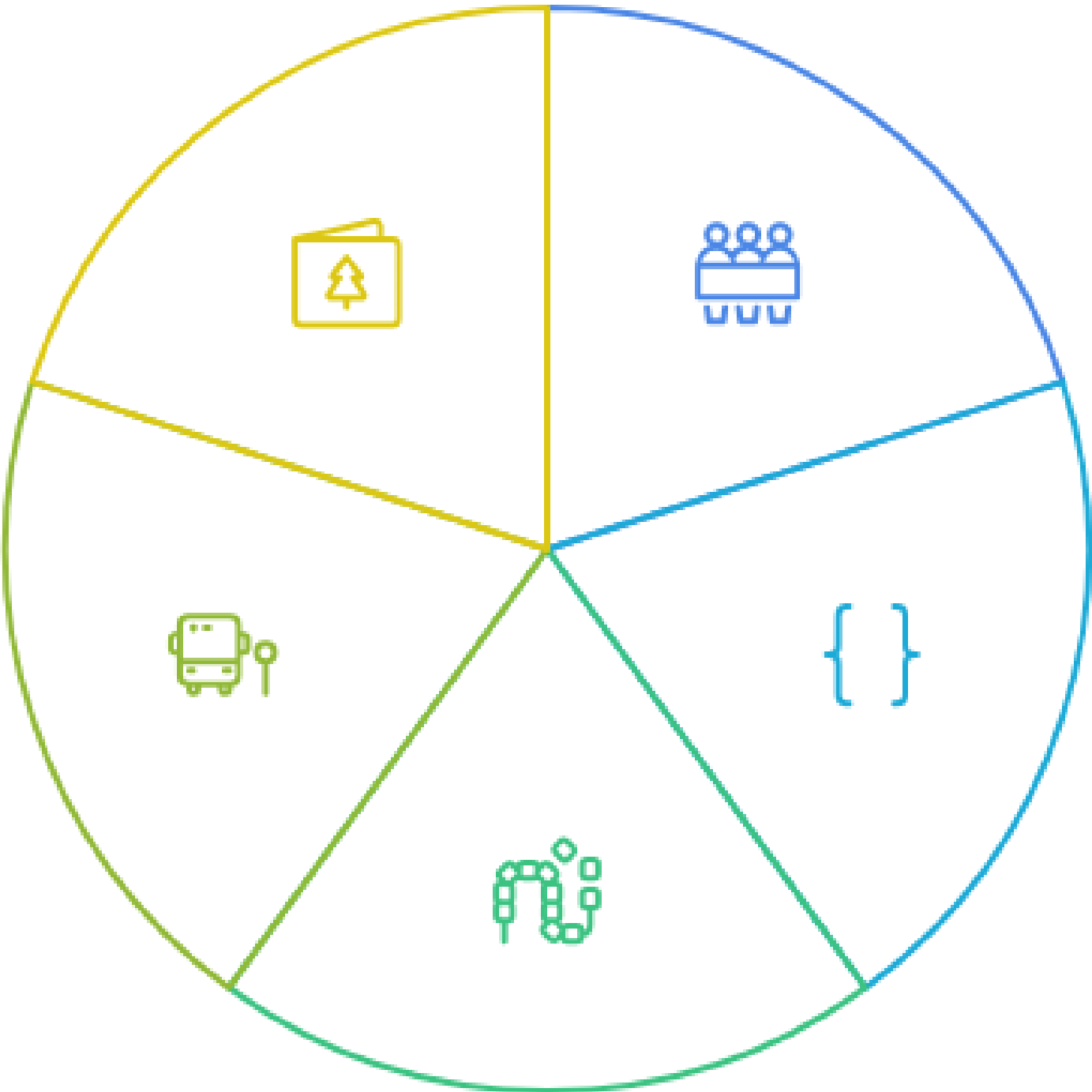
System Architecture & Class Design

Layer	Components
 Layer 1 – Presentation Layer	TransportGUI.java – 8-panel Swing GUI, dark-themed, sidebar navigation
 Layer 2 – Business Logic Layer	Main.java – Controller for all CRUD, allocation, search, sort, validation, reporting
 Layer 3 – Entity Layer	Bus, Driver, Student, Route, Stop, TransportationException
 Relationships	Bus holds Student[], Bus assigned one Driver, Route contains LinkedList<Stop>, Student allocated to Bus



DATA STRUCTURE USED

Distribution of Data Structure Applications (applications)



- 1 Array
Student seating slots
- 1 ArrayList
Buses, Students, Drivers
- 1 LinkedList
Route stops, Travel history
- 1 HashMap
Bus number to Bus object mapping
- 1 TreeMap
Route number to Route mapping

Complexity callout

	10^x Complexity
Bus Lookup	O(1)
Route Access	O(log n)
Sort Buses	O(n log n)
History Append	O(1)



Key Features & Conclusion

Module	 Student Management	 Bus Management	 Driver Management	 Route Management	 Stop Management	 Student-Bus Allocation	 Search & Sort	 History & Reports
Key Highlights	Full CRUD for all 5 entities	Search by Student ID, Name, Bus No., Route No.	Sort buses by capacity	Routes auto-sorted via TreeMap	Atomic allocation – capacity checked before removing student	Custom TransportationException for domain-level error handling	Click-to-prefill tables – no manual re-entry	Pre-loaded demo data (5 buses, drivers, students, routes, stops)

Demonstrates practical application of Java data structures, OOP, exception handling, and GUI programming to solve a real-world school transportation problem.



OUTPUTS

BusTrack Pro

School Transportation Management System

SYSTEM ACTIVE

NAVIGATION

Students

Buses

Drivers

Routes

Stops

Allocation

Search & Sort

History & Report

Students

Add, update, delete and view student records

STUDENT ID

FULL NAME

AGE

1006

Priya Sharma

15

CLASS

ADDRESS

10-A

14, Rose Villa

+ Add Student

Update

Delete

Refresh

ID	Name	Age	Class	Address	Bus
1001	Aarav Sharma	14	9-A	12, Lal Bagh	Bus #1
1002	Diya Patel	13	8-B	45, Rose Garden	Bus #2
1003	Rohan Verma	15	10-A	78, Sector 5	Bus #3
1004	Sneha Reddy	12	7-C	23, Green Park	Bus #4
1005	Arjun Nair	16	11-B	90, Hill View	Bus #5
1006	Priya Sharma	15	10-A	14, Rose Villa	Bus #6

v1.0 - School Bus System

Java Swing - School Bus Tracking System

Added: Priya Sharma (ID: 1006)

BusTrack Pro

School Transportation Management System

SYSTEM ACTIVE

NAVIGATION

Students

Buses

Drivers

Routes

Stops

Allocation

Search & Sort

History & Report

Buses

Manage bus fleet, capacity and driver assignments

BUS NUMBER

BUS NAME

CAPACITY

6

Silver Express

50

+ Add Bus

Update Name

Delete

Refresh

Bus #	Name	Capacity	Passengers	Driver
1	Blue Express	40	1 / 40	Ramesh Kumar
2	Red Rocket	35	1 / 35	Suresh Yadav
3	Green Shuttle	30	1 / 30	Maresh Singh
4	Yellow Star	45	1 / 45	Dinesh Sharma
5	Orange Cruiser	25	1 / 25	Ganesh Patel
6	Silver Express	50	1 / 50	Vikram Malhotra

v1.0 - School Bus System

Java Swing - School Bus Tracking System

Added bus #6 — Silver Express

BusTrack Pro

School Transportation Management System

SYSTEM ACTIVE

NAVIGATION

Students

Buses

Drivers

Routes

Stops

Allocation

Search & Sort

History & Report

Drivers

Manage drivers, contact info and bus assignments

DRIVER ID

FULL NAME

PHONE NUMBER

LICENSE NUMBER

106

Vikram Malhotra

9811223344

DL-106-UP

+ Add Driver

Update

Delete

Assign to Bus...

Refresh

Driver ID	Name	Phone	License
101	Ramesh Kumar	9876543210	DL-101-MH
102	Suresh Yadav	9123456780	DL-102-DL
103	Maresh Singh	9988776655	DL-103-UP
104	Dinesh Sharma	9001122334	DL-104-RJ
105	Ganesh Patel	9876001234	DL-105-CJ
106	Vikram Malhotra	9811223344	DL-106-UP

v1.0 - School Bus System

Java Swing - School Bus Tracking System

Added driver: Vikram Malhotra (ID: 106) → Assigned to Bus #6

BusTrack Pro

School Transportation Management System

SYSTEM ACTIVE

NAVIGATION

Students

Buses

Drivers

Routes

Stops

Allocation

Search & Sort

History & Report

Routes

Define transport routes and manage stop assignments

ROUTE NUMBER

ROUTE NAME

6

Express Highway Route

+ Add Route

Update

Delete

Sort by Number

Refresh

Route #	Name	Stops
1	North Route	1
2	South Route	1
3	East Route	1
4	West Route	1
5	Central Route	1
6	Express Highway Route	2

v1.0 - School Bus System

Java Swing - School Bus Tracking System

Added route #6: Express Highway Route

BusTrack Pro

School Transportation Management System

SYSTEM ACTIVE

NAVIGATION

Students

Buses

Drivers

Routes

Stops

Allocation

Search & Sort

History & Report

Stops

Manage bus stops and assign them to routes

STOP ID

STOP NAME

LOCATION

306

Tech Park Phase 1

Sector 62

+ Add Stop

Delete

Add to Route...

Remove from Route

Refresh

Stop ID	Name	Location
301	City Center	MG Road
302	Railway Station	Station Road
303	Market Square	Nehru Place
304	Park Colony	Sector 14
305	Mall Junction	Ring Road
306	Tech Park Phase 1	Sector 62

v1.0 - School Bus System

Java Swing - School Bus Tracking System

Added stop #306: Tech Park Phase 1 (Assigned to Route #6)

BusTrack Pro

School Transportation Management System

SYSTEM ACTIVE

NAVIGATION

Students

Buses

Drivers

Routes

Stops

Allocation

Search & Sort

History & Report

Allocation

Assign students to buses with automatic capacity checking

STUDENT ID

BUS NUMBER

1006

6

✓ Allocated Successfully!

Student: Priya Sharma (ID: 1006)

Bus: Silver Express (Bus #6)

Seats: 1 / 50 | Route #6

Allocate Student to Bus...

v1.0 - School Bus System

Java Swing - School Bus Tracking System

Allocated student 1006 to bus #6





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Thank You